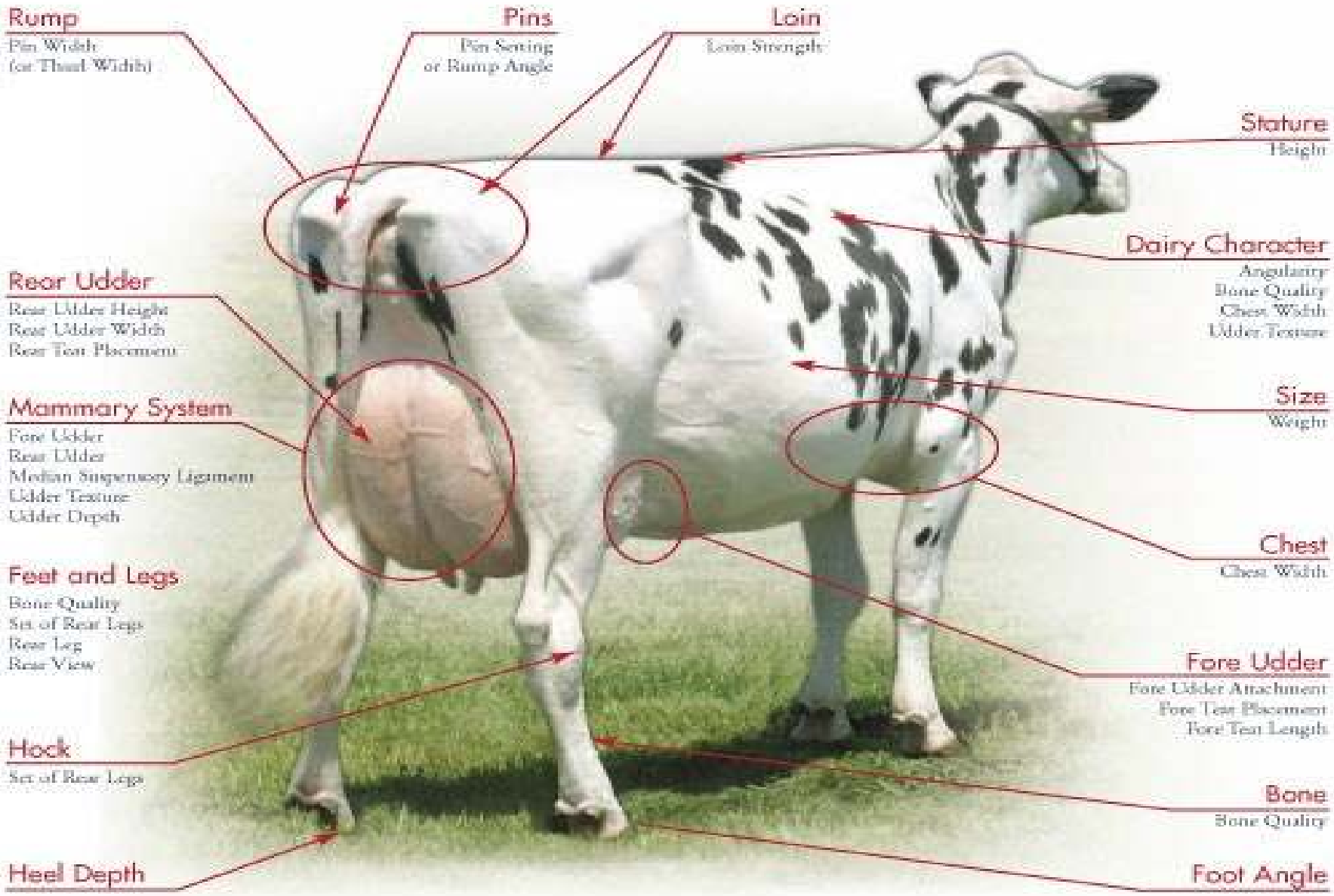
乳牛乳房結構與泌乳生理

主講者：新竹分所 李國華
資料來源：產業組 蕭宗法
99 乳牛乳房炎防治講習班






Rump
Pin Width
(or Thurl Width)
Pins
Pin Setting
or Rump Angle
Loin
Loin Strength
Stature
Height
Dairy Character
Angularity
Bone Quality
Chest Width
Udder Texture
Rear Udder
Rear Udder Height
Rear Udder Width
Rear Teat Placement
Size
Weight
Mammary System
Fore Udder
Rear Udder
Median Suspensory Ligament
Udder Texture
Udder Depth
Chest
Chest Width
Feet and Legs
Bone Quality
Set of Rear Legs
Rear Leg
Rear View
Fore Udder
Fore Udder Attachment
Fore Teat Placement
Fore Teat Length
Hock
Set of Rear Legs
Bone
Bone Quality
Heel Depth
Foot Angle

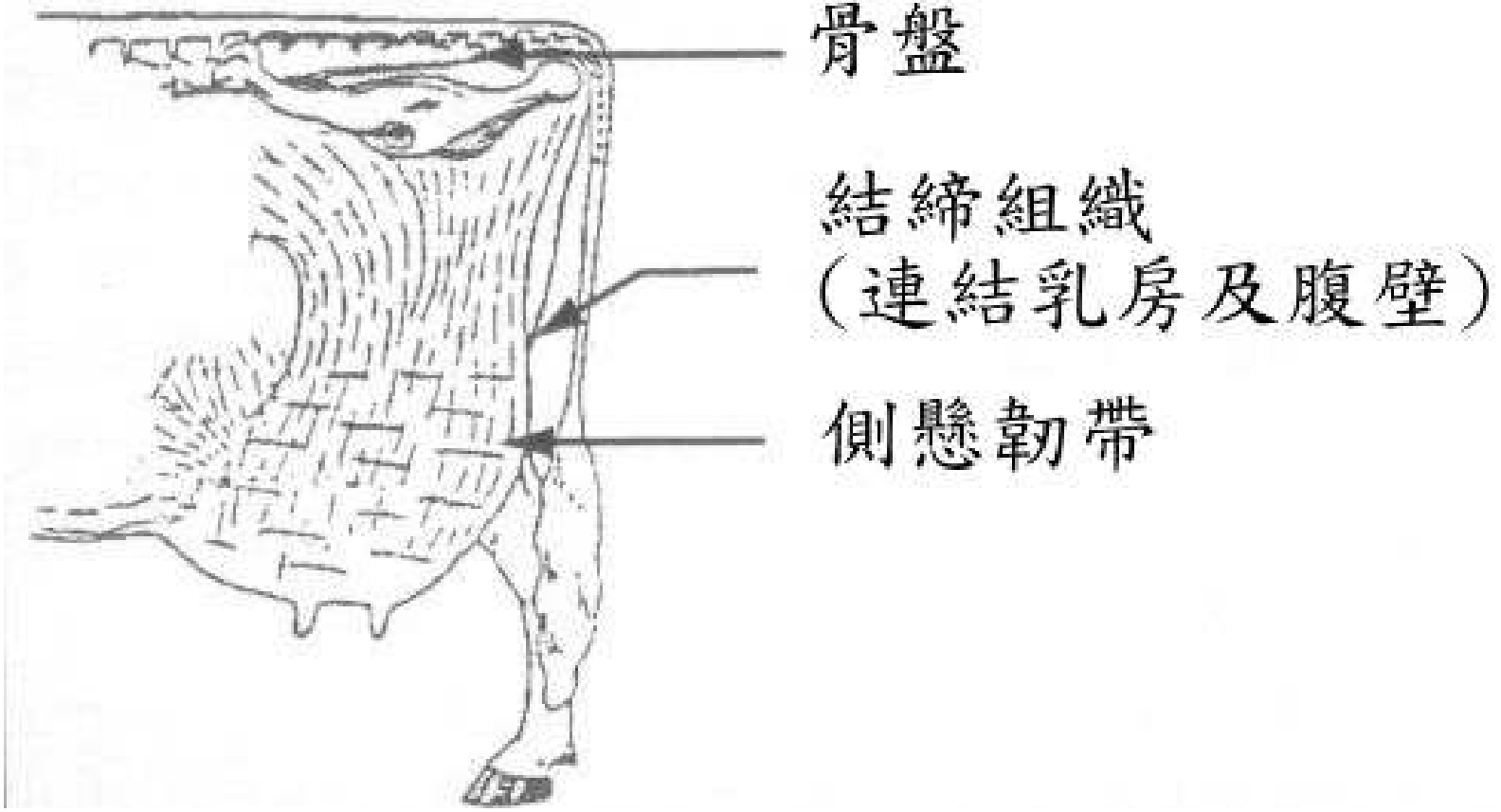
乳房

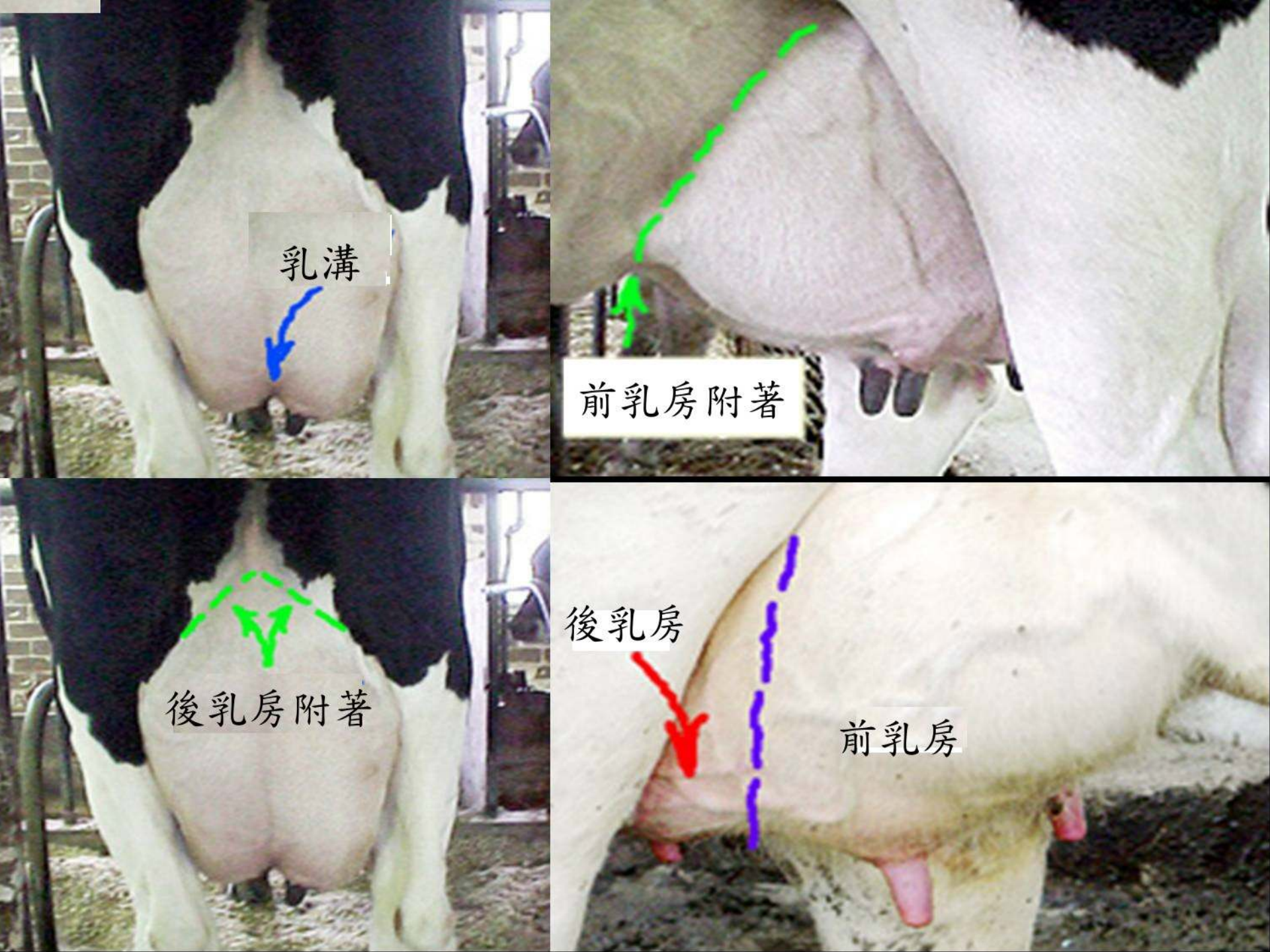


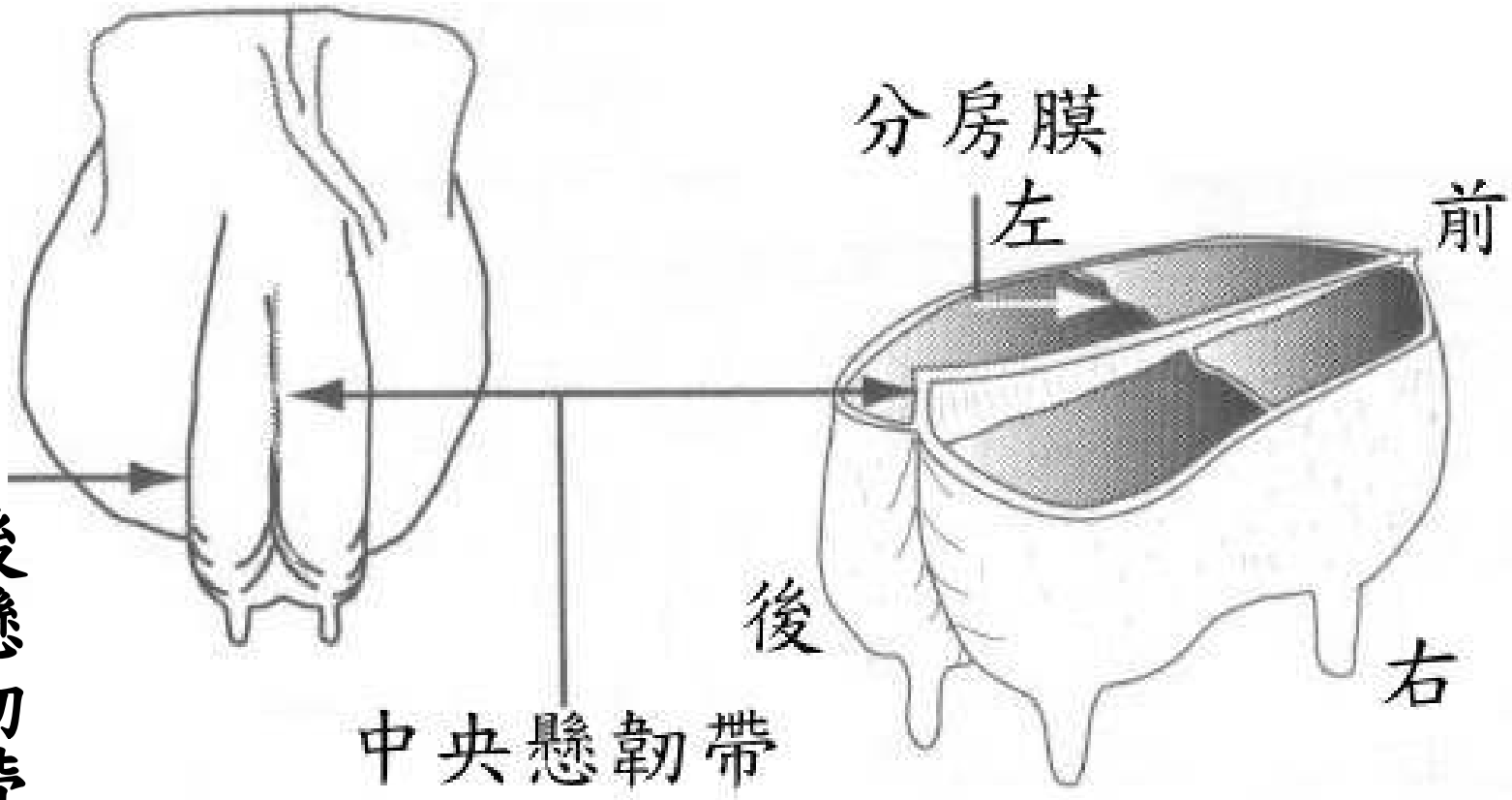
乳房是一個完整功能的器官，包括

- 支持系統
- 分泌系統
- 乳管系統
- 血液、淋巴與神經系統

1. 乳房支持系統



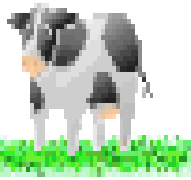

乳溝
前乳房附著
後乳房附著
後乳房
前乳房


分房膜
左
前
後
右
後懸靭帯
中央懸靭帯

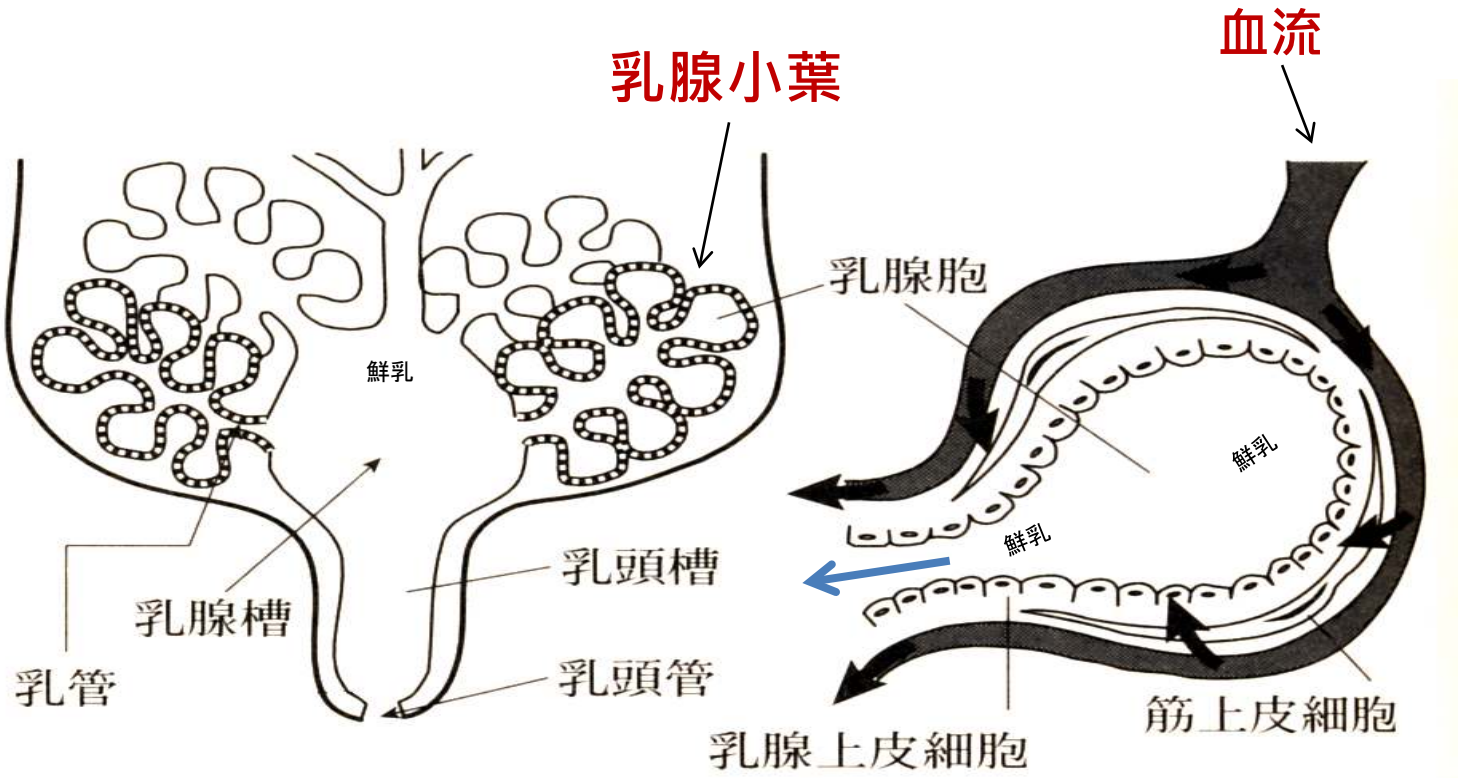
2.分泌系統

- 乳腺是牛乳分泌的結構，包括
  (1) 乳泡
  (2) 管道系統
  (3) 乳頭

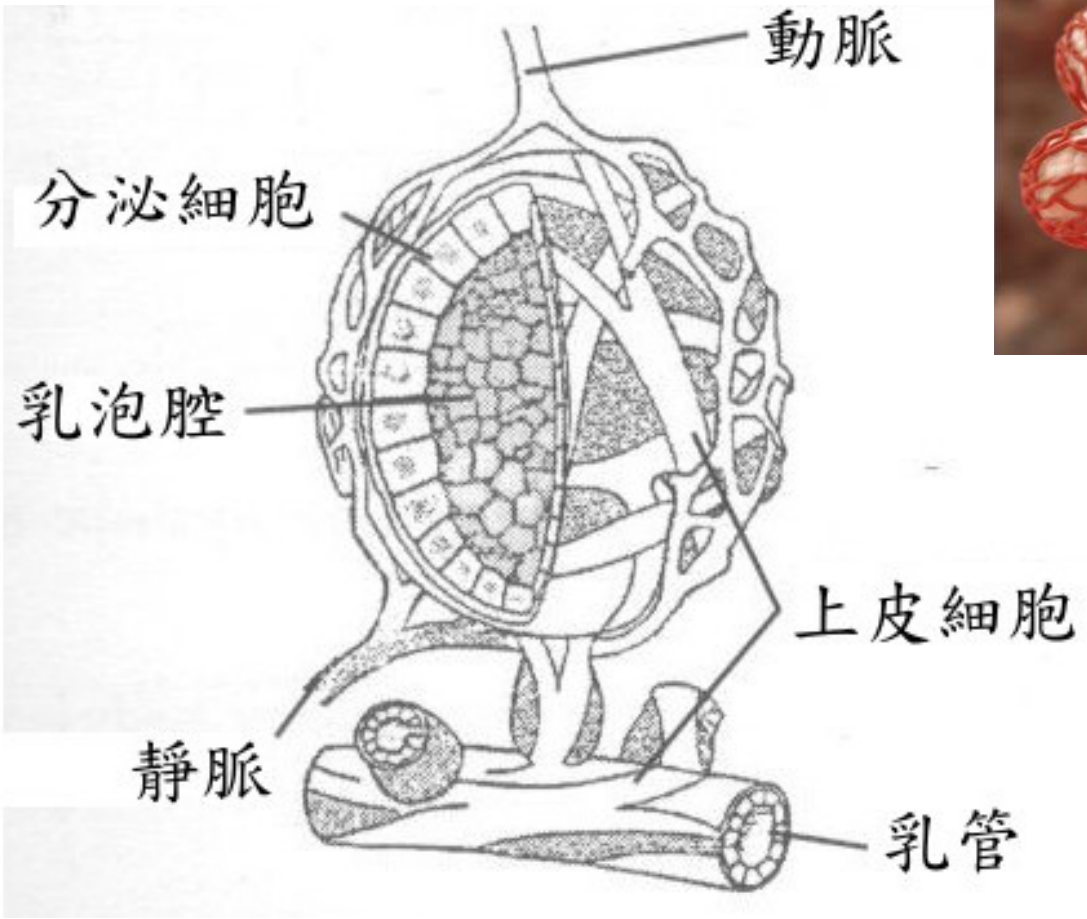




要生產 1 公斤的牛乳必須
要 400～500 公斤的血液



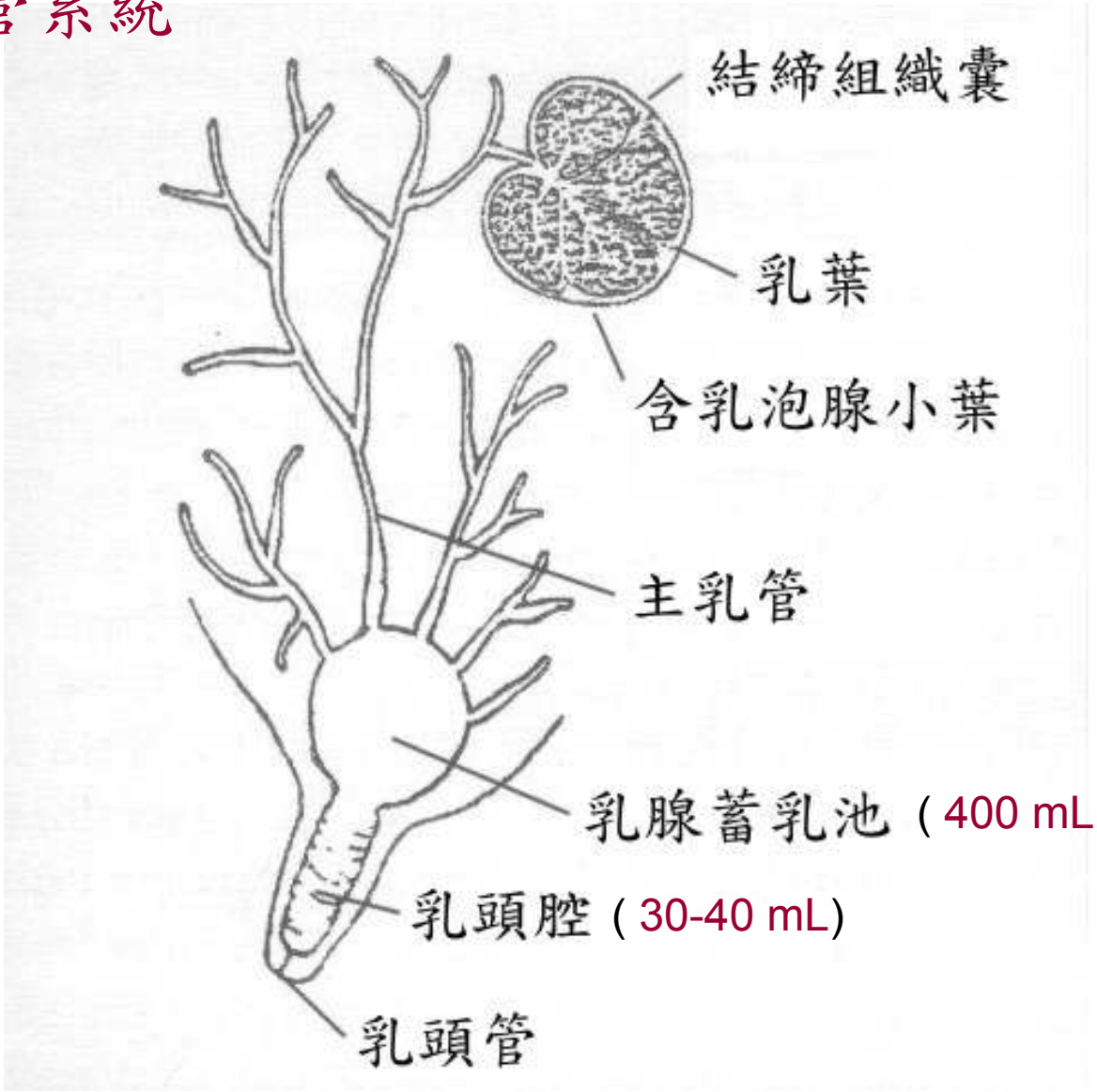
(1) 乳腺泡結構



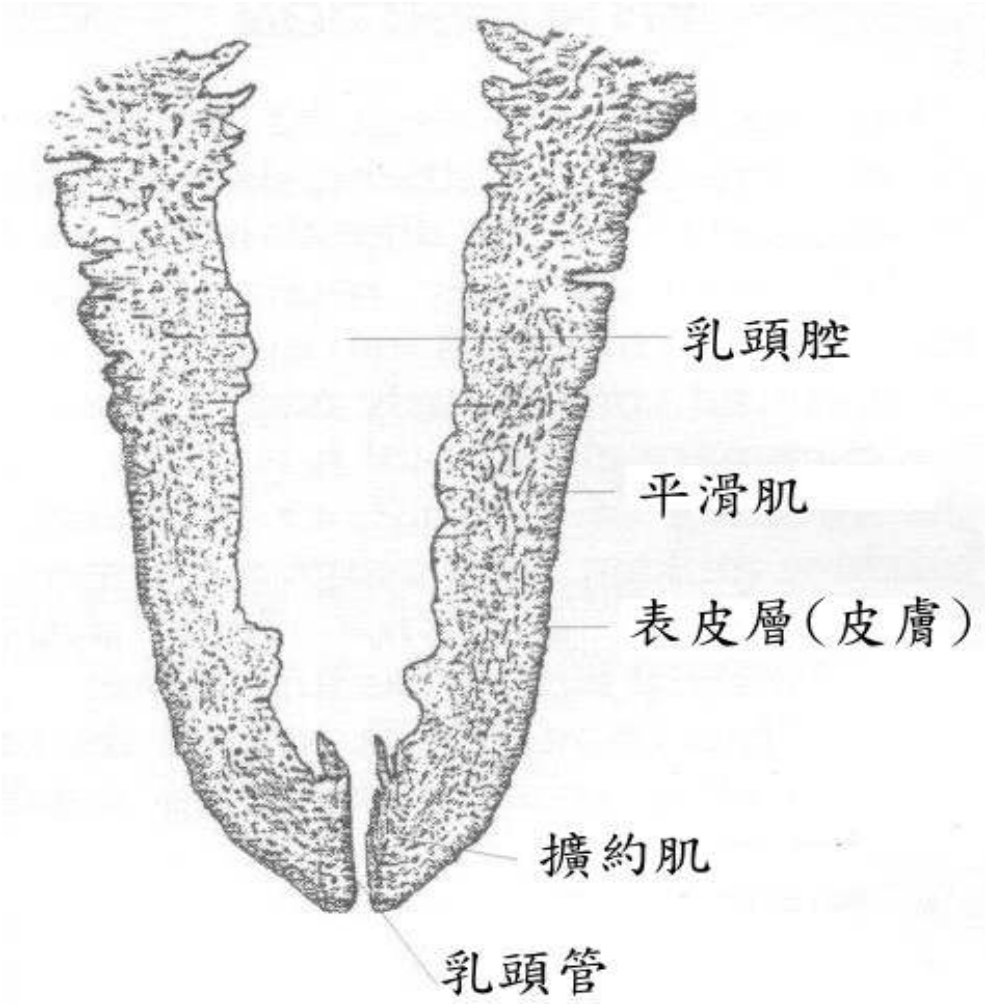




(2) 乳管系統



(3) 乳頭結構



The teat canal

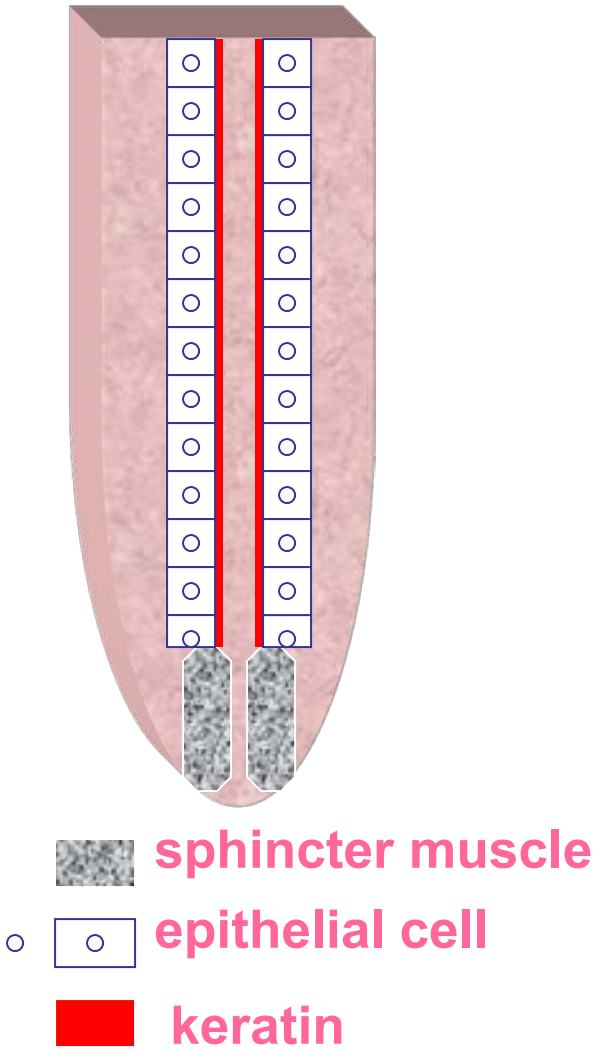
括約肌

Keep the orifice closed and thereby isolate the interior of the MG

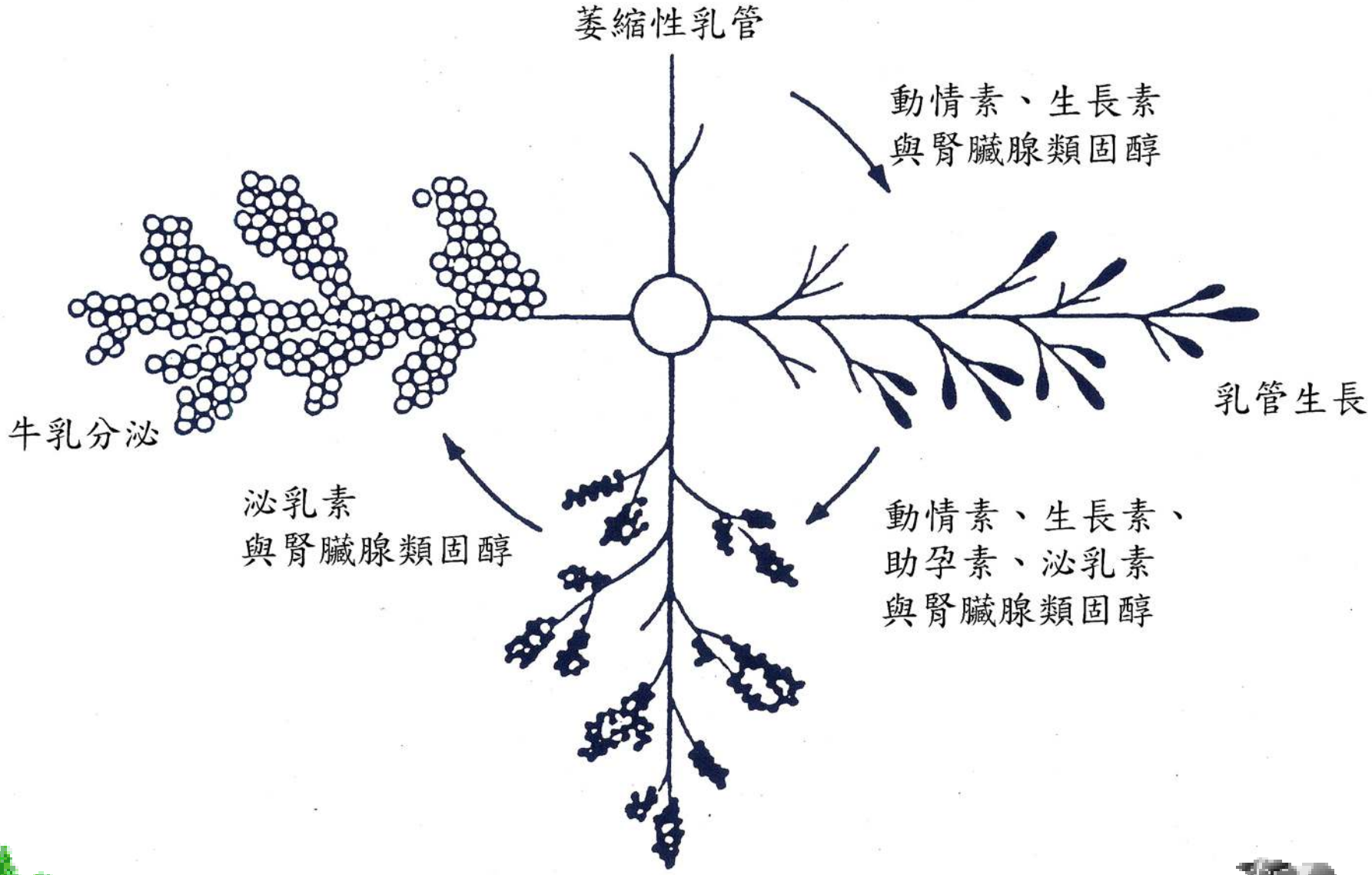
角質層(Keratin)

A waxy material traps invading bacteria

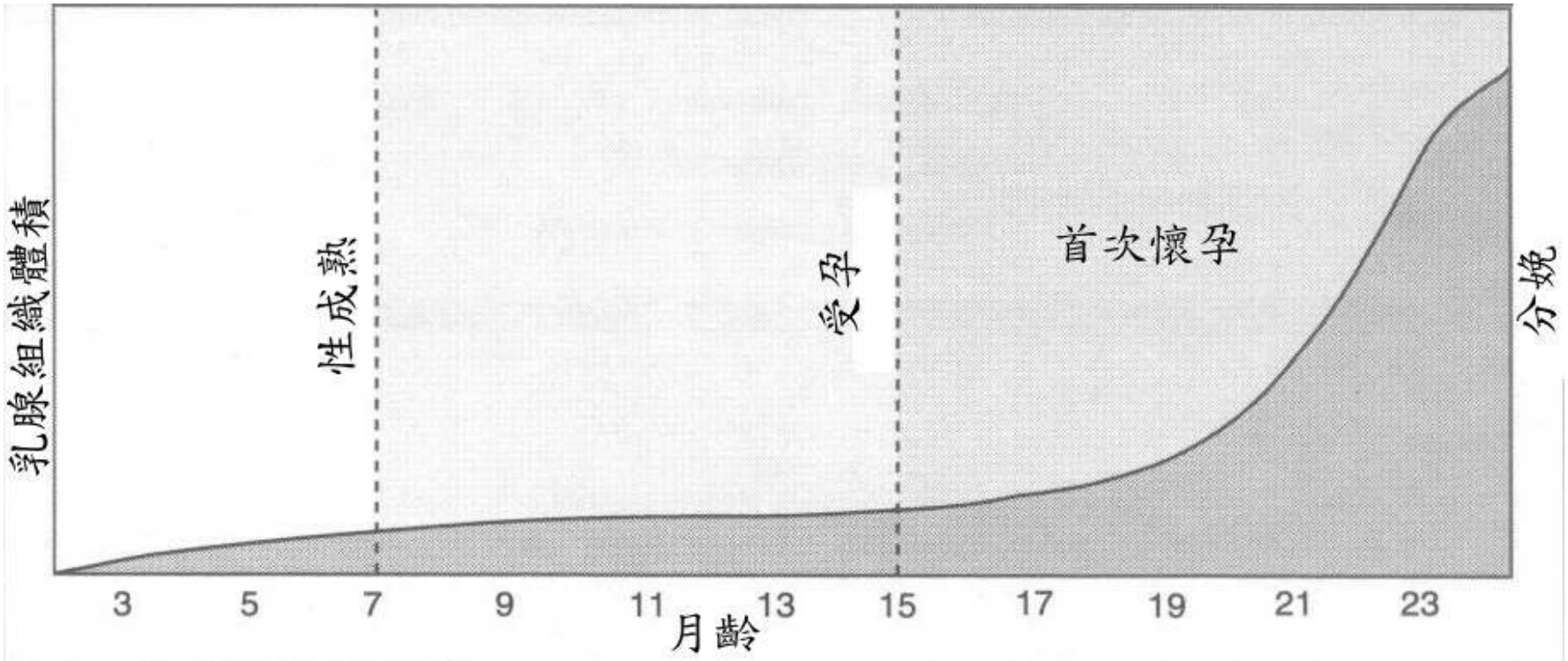
and prevents bacteria from migrating into the milk cistern



3. 乳腺的發育

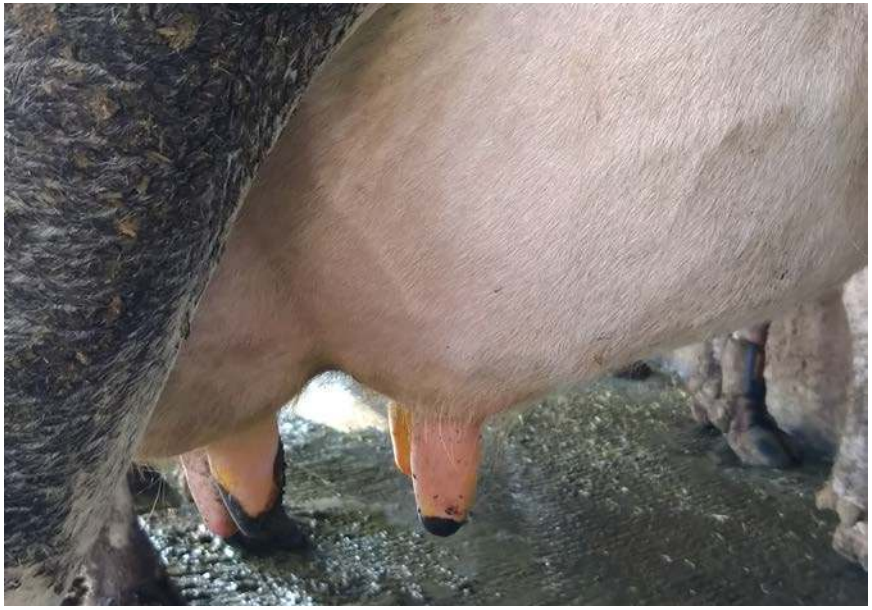

萎縮性乳管
動情素、生長素
與腎臟腺類固醇
乳管生長
牛乳分泌
泌乳素
與腎臟腺類固醇
動情素、生長素、
助孕素、泌乳素
與腎臟腺類固醇
乳小葉-乳腺泡生長

乳腺組織之生長與發育



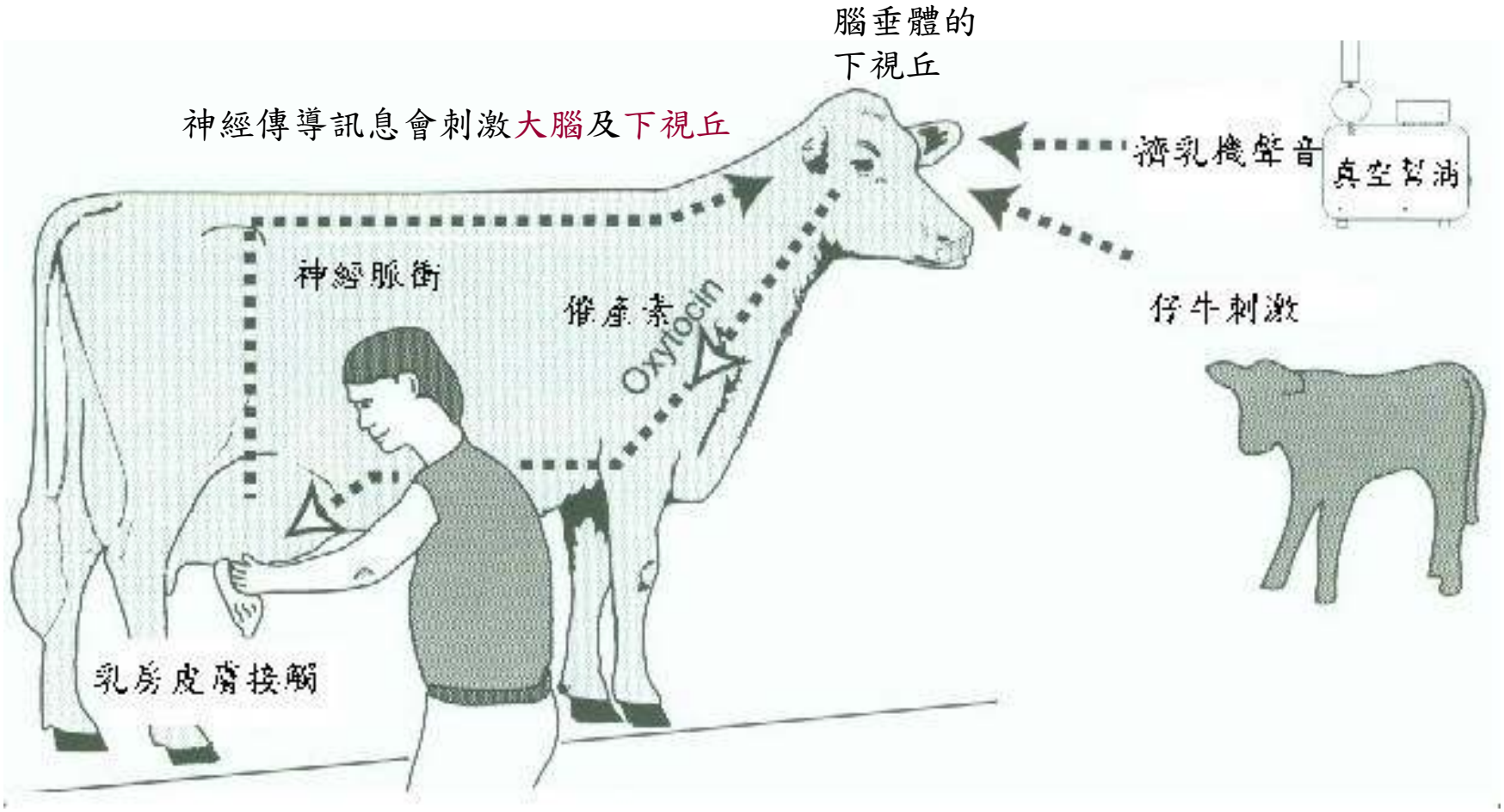






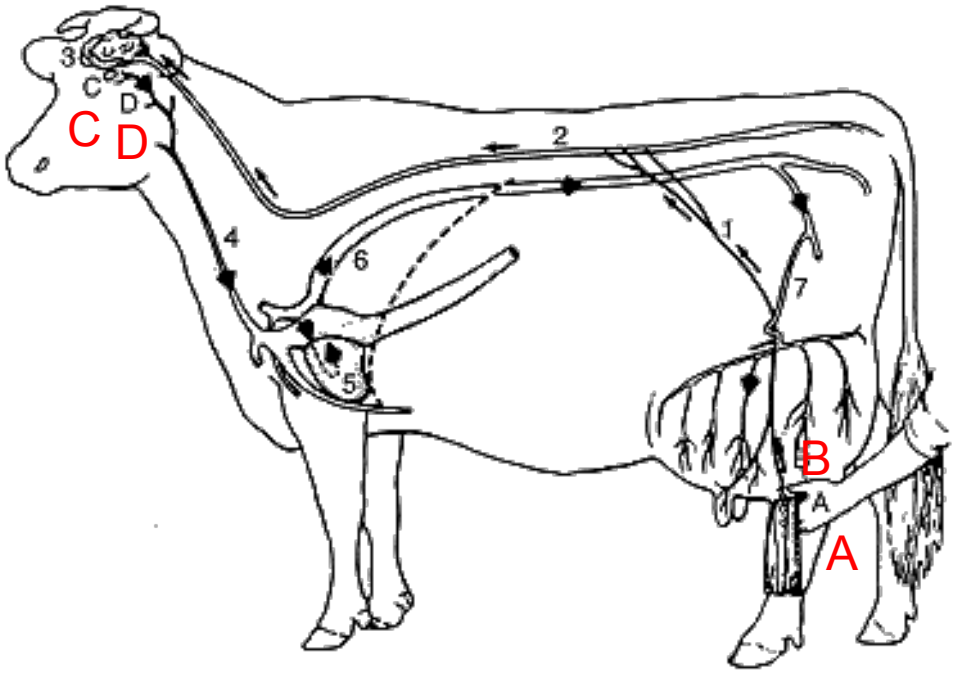


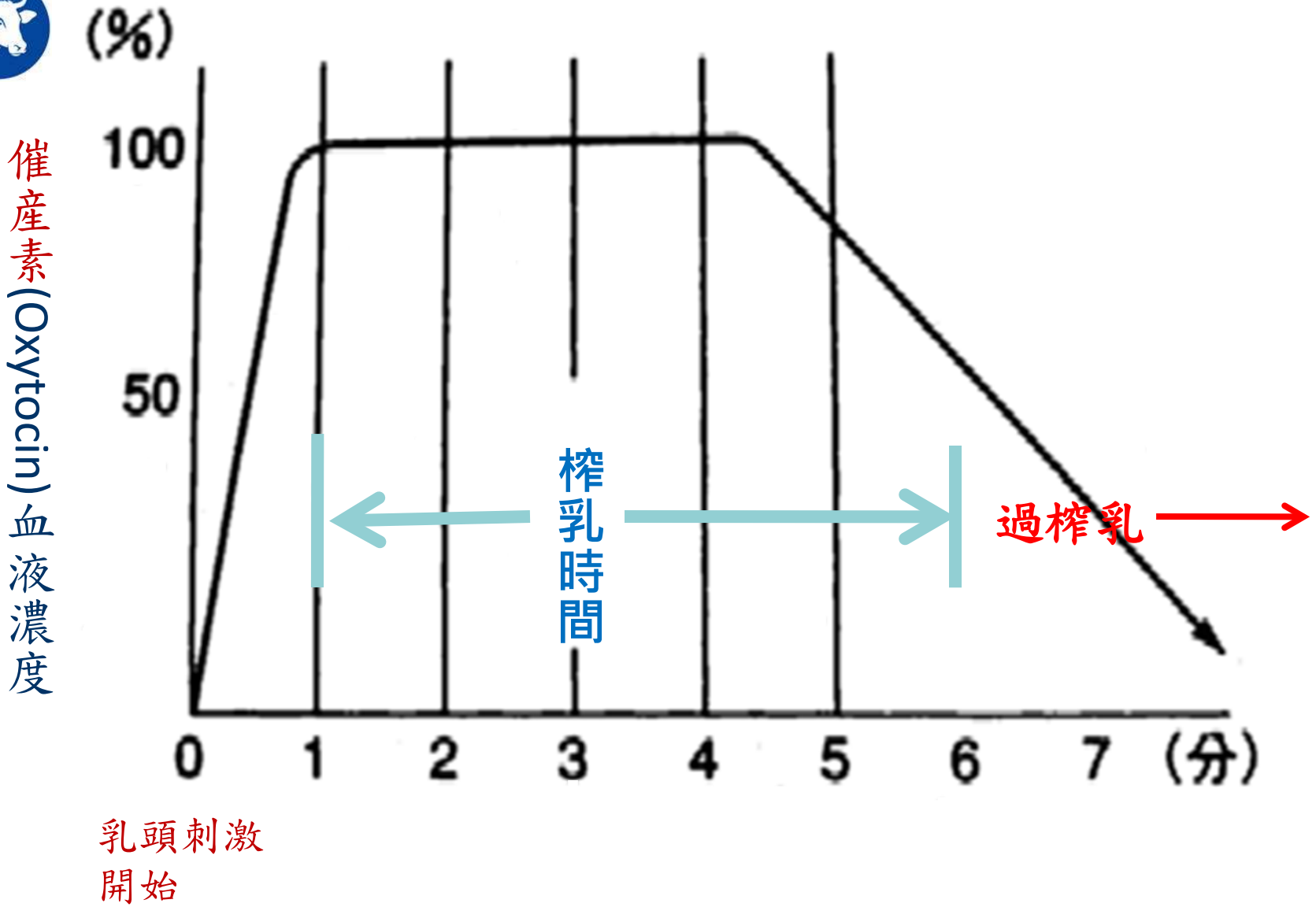
4. 泌乳神經及內泌素調控



泌乳神經及內泌素調控

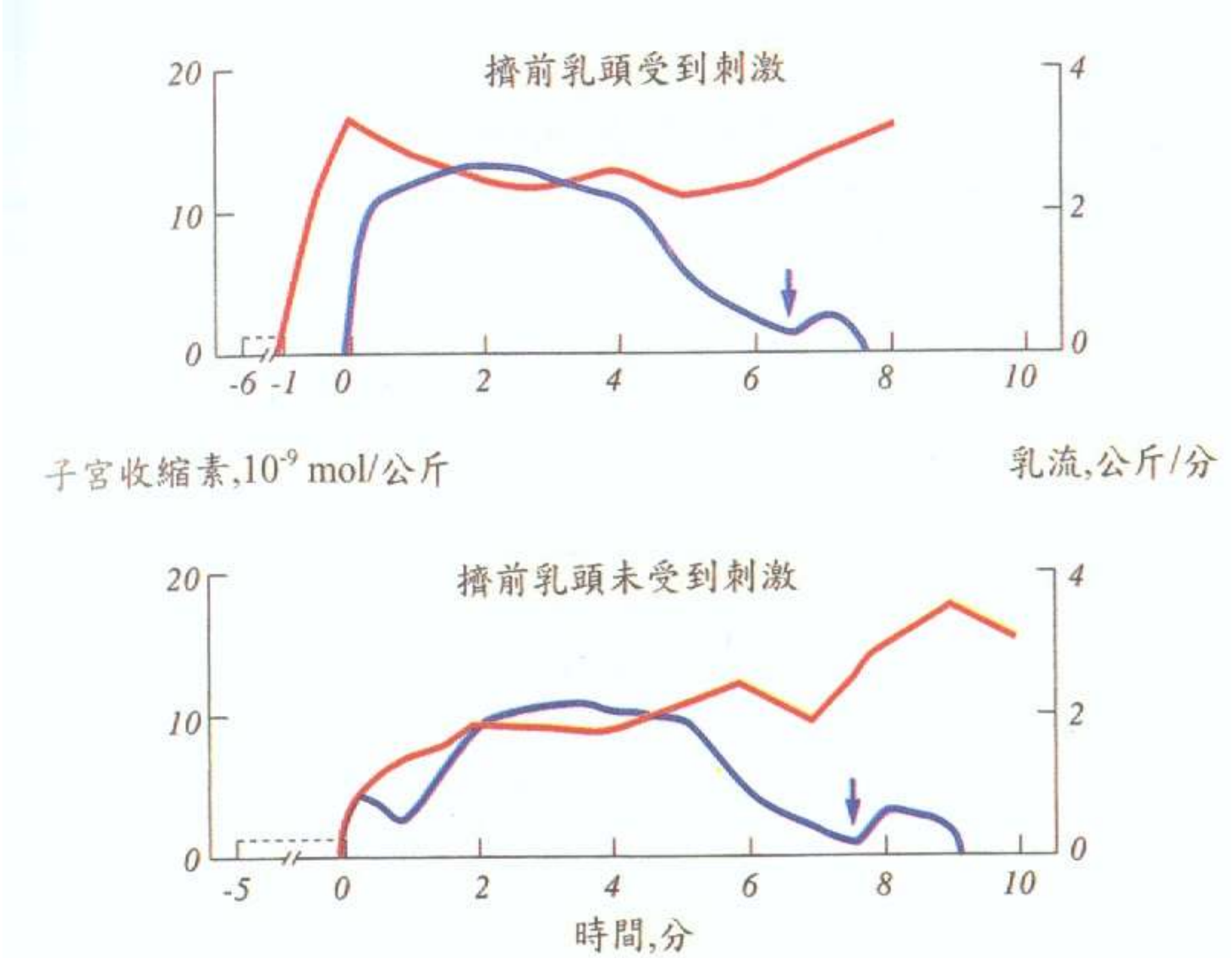
- 乳房皮膚接觸的刺激 (A)，造成神經脈衝 (B) 經由鼠蹊部神經 (1) 傳導到脊髓 (2) 及腦 (3)。
- 腦接到刺激由腦垂體後葉 (D) 分泌催產素 (C) 。催產素釋出後進入頸靜脈分枝 (4) 及心臟 (5) ，隨後經由動脈血傳遞訊息到全身各處。
- 催產素經由主動脈 (6) 離開心，再經由外陰動脈 (7)) 進入到乳房。
- 催產素在乳房造成肌上皮細胞收縮，使乳汁由乳腺泡釋出。





從乳頭刺激（前榨）開始，催産素(oxytocin)濃度的變化

擠乳刺激與泌乳的關係：藍線為乳流、紅線為催產素濃度



泌乳生理

- 1 公升乳須 400~500公升血液來供應
- 乳汁貯留在乳腺胞、乳小管內、大乳管、乳腺槽部
- 乳腺肌上皮細胞因 oxytocin 分泌而收縮
- **Oxytocin** 分泌持續時間約 20~40 秒
  作用持續時間 2~10 分鐘

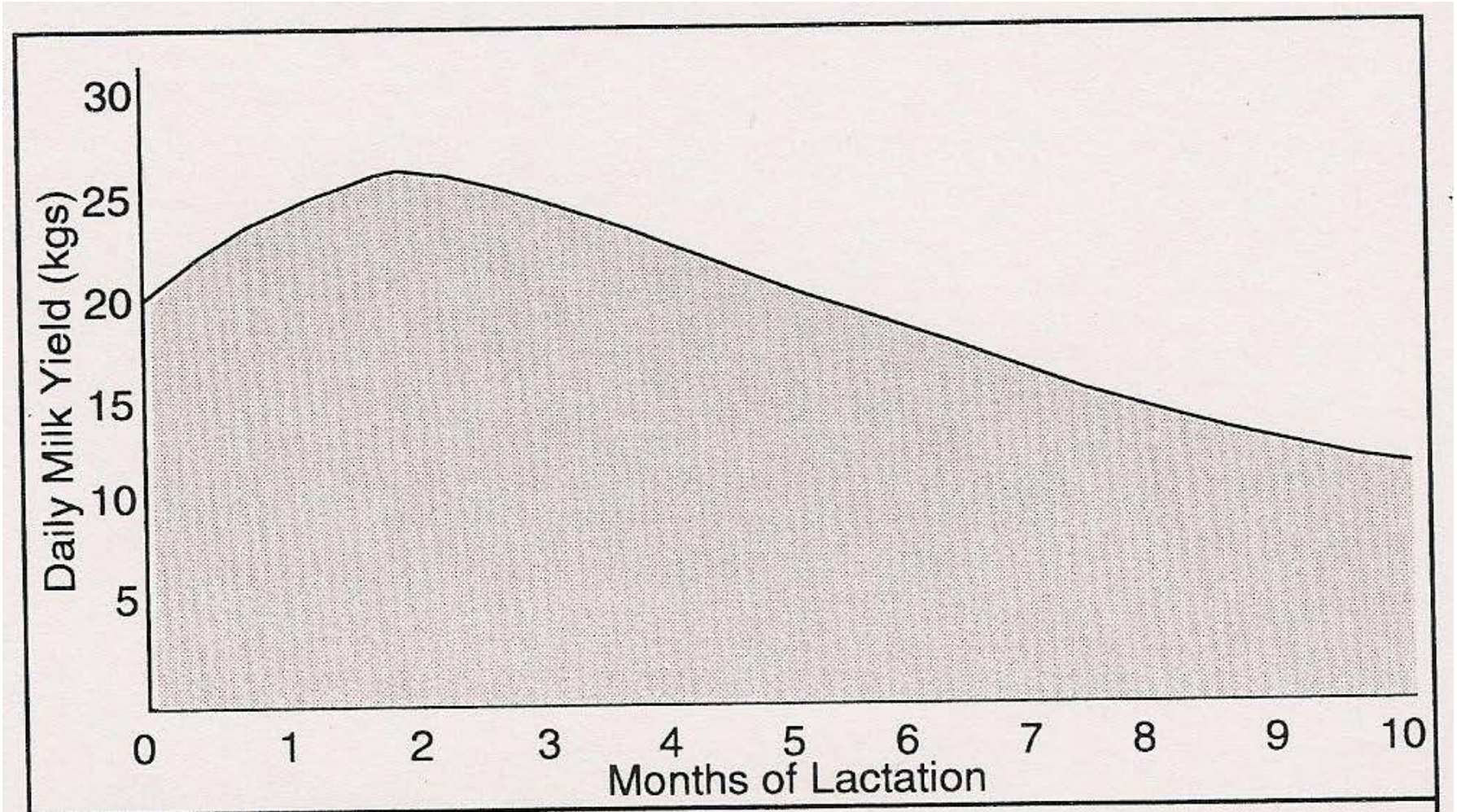


Figure 4.2: Typical lactation curve

泌乳曲線